

BINISHA NAGA

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SUMMARY

Full Stack Developer with hands-on production experience across React, Next.js, FastAPI, and AWS. Builds end-to-end features from pixel-perfect interfaces to deployed cloud infrastructure.. Actively exploring AI/ML integration in practical applications.

WORK EXPERIENCE

Full Stack Developer

06/2025 – Present

Blys

- Maintain and ship features across four interconnected products(Web App, Pro App, Client Mobile App, and Admin Panel) built on React, React Native and Node.js/Express.
- Work across the full stack: designing MySQL schemas and REST APIs on the backend, and building polished TypeScript/React UIs on the frontend.
- Build and maintain customer communication flows including SMS notifications, email templates, push notifications and worked on booking engine.
- Collaborate with design and QA to scope requirements, triage bugs, and deliver releases on schedule.

Technologies: React, React Native, Node.js, JavaScript, TypeScript, mySQL

Full Stack Developer

07/2024 – 06/2025

Dhimay Tech

- Engineered responsive, accessible UIs with React, Next.js, TypeScript, and Tailwind CSS; integrated Framer Motion animations and shadcn/Material Tailwind component libraries.
- Built complex form systems with React Hook Form; managed server state via TanStack Query and global state via Zustand.
- Designed and delivered RESTful APIs with FastAPI covering auth (JWT/OAuth2), dynamic content, bookings, media uploads, and admin operations.
- Architected and optimized PostgreSQL schemas with SQLAlchemy; wrote unit and integration tests with pytest covering API endpoints, authentication flows, and business logic across all backend services.
- Wrote frontend unit tests to validate component behavior, form logic, and state management.
- Provisioned and maintained AWS infrastructure: EC2 (Docker + bare-metal), RDS PostgreSQL, S3 (media + static hosting), Amplify, and SES with resource tagging for cost accountability.

Technologies: React, FastAPI, Python, AWS, TypeScript, TailwindCSS, PostgreSQL, Zustand, flutter

EDUCATION

Bachelor in Computer Science and Information Technology

2020 – 2025

Samriddhi College

+2 (Science)

2018 – 2020

Khwopa Secondary School

SKILLS

- | | |
|----------------|----------------|
| • React | • Next.js |
| • FastAPI | • TypeScript |
| • Tailwind CSS | • React Native |
| • AWS | • Python |

SELF PROJECTS

Ascend

Web: <https://ascend.binishanaga.com.np/> Android: <http://downloads.binishanaga.com.np/ascend.apk>

- Built a full-stack productivity platform featuring hierarchical goal tracking, calendar-based task scheduling, vision boards, rich note-taking and affirmations audio play.
- Architected a serverless Python/FastAPI backend deployed on AWS Lambda with async PostgreSQL via SQLAlchemy and Alembic migrations.
- Implemented multi-method authentication: Google OAuth, JWT sessions, bcrypt password hashing, and email OTP verification.
- Integrated AWS S3 for vision board image uploads and affirmation audio storage and playback.
- Deployed frontend on Vercel and managed all infrastructure configuration for a production-ready, scalable architecture.

Tech Stack

- frontend: Next.js, TypeScript, Tailwind CSS v4, Zustand, TanStack Query, Radix UI
- Backend: FastAPI, Python
- Database: PostgreSQL, Supabase
- Deployment: AWS Lambda, S3, Vercel

Chess

Web Link: <https://chess.binishanaga.com.np/> Android: <http://downloads.binishanaga.com.np/chess.apk>

- Built cross-platform chess app from a single codebase; all game logic, UI, and animations shared across platforms
- Implemented 4 game modes: local, vs AI, 29-lesson interactive learn mode, and real-time online multiplayer
- Integrated on-device Stockfish engine via native FFI with 8 difficulty levels; used softmax temperature sampling to reach ~400 Elo (same technique as Lichess bots)
- Designed serverless AWS backend: AppSync GraphQL + WebSocket subscriptions, Lambda functions for server-authoritative move validation, DynamoDB, and Cognito guest auth.
- Implemented optimistic UI for online moves with server reconciliation and clean rollback on rejection.
- Built full in-game action suite: resign, offer/accept/decline draw, request/accept/decline takeback, rematch.
- Added in-game coach with real-time eval bar, blunder detection, hints, and post-game per-move quality grading (brilliant → blunder) with accuracy score
- Wrote unit tests covering game logic and online state machine using in-memory fakes (no AWS calls in tests)

Tech Stack

- Flutter/ Dart, AWS, Riverpod

Sudoku

Web Link: <https://sudoku.binishanaga.com.np/> Android: <http://downloads.binishanaga.com.np/sudoku.apk>

- Cross-platform game targeting Android & Web with React Native + react-native-web + Expo Router file-based navigation
- Implemented custom 2D canvas rendering via @shopify/react-native-skia for smooth board drawing and cell animations
- Designed offline-first architecture using react-native-mmkv (native key-value store) for persistent game state, stats, and settings
- Managed complex game state across puzzle, settings, and achievements with Zustand v5 stores
- Solved UI-freezing puzzle generation by pre-generating a bundled puzzle bank at build time (Node.js script), eliminating 3–11s synchronous on-device computation
- Integrated Google Mobile Ads (AdMob) with react-native-google-mobile-ads for monetization
- Added haptic feedback + audio via expo-haptics and expo-audio for tactile game feel
- Configured EAS Build pipeline for Android release APKs with OTA update support via expo-updates
- Written in TypeScript 6 with Jest + jest-expo test suite for game logic (validator, scorer, puzzle engine)

Tech Stack

- React, React Native, Typescript, Zustand, Jest